

Effect of LLM Temperature on Rule-Grounded Response Quality

A Randomized Complete Block Design Experiment

By Shannon O'Donovan shannon.odonovan@terra-cog.com

Design of Experiments • Anthropic • Building with Claude API • Claude Code • February 2026

1. Problem Statement

Retrieval-Augmented Generation (RAG) is increasingly used to build domain-specific AI agents that answer questions from an authoritative source rather than from the model's training data alone.^{1,2} A critical open question in deploying such systems is whether the model's temperature parameter—which controls the randomness of token sampling—materially affects the quality of the responses it produces.^{3,4}

Known challenges of LLMs include:

- Presenting false information when it does not have the answer.
- Presenting out-of-date or generic information when the user expects a specific, current response.
- Creating a response from non-authoritative sources.
- Creating inaccurate responses due to terminology confusion, wherein different training sources use the same terminology to talk about different things.⁵

To address this, I designed a RAG-based teaching agent to answer questions about the board game Clue as a surrogate for production systems where its agent's response is grounded in the authoritative documents, not the LLM training model. Retrieval-augmented generation systems are governed by multiple interacting factors — including chunking strategy, retrieval depth, prompt structure, and model behavior. In practice, performance differences are often discussed qualitatively or attributed to individual components without controlled comparison.

The agent uses a hybrid retriever (BM25 + vector similarity with contextual reranking), a defined system prompt restricting responses to rule-grounded answers, and explicit citation requirements. The game rulebook provides a fixed, unambiguous specification against which all responses can be objectively scored.

¹ Lewis et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks"; Amaz. Web Serv. Inc, "What Is RAG?"

² Lewis et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks."

³ Renze and Guven, "The Effect of Sampling Temperature on Problem Solving in Large Language Models."

⁴ Note - prompt was held as a constant

⁵ Amaz. Web Serv. Inc, "What Is RAG?"

C4 Context Diagram of Solution



The practical question motivating the study is: ***Does running the model at a higher temperature (more random) produce meaningfully worse rule-grounded answers than running it at a lower temperature (more deterministic), after accounting for the inherent variation in question difficulty?***

If temperature has no material effect, practitioners can simplify deployment decisions and tune the parameter freely for other system objectives. If temperature does matter, it becomes a configuration factor requiring deliberate calibration.

2. Experimental Factors

2.1 Treatment Factor: Temperature

Temperature was varied across three pre-specified levels representing low (more deterministic), medium, and high (more stochastic) sampling behavior. These levels were chosen to span a practically meaningful range without entering the regime where output coherence degrades.⁶

Low: $T = 0.3$ (near-deterministic sampling)
Medium: $T = 0.5$ (baseline / conventional default)
High: $T = 0.8$ (noticeably increased randomness)

2.2 Blocking Factor: Question Type

Questions are not equal in difficulty. A straightforward factual question (“How many players can play Clue?”) will produce a high-quality answer at any temperature. A strategically misleading or out-of-scope question places heavier demands on the agent’s scope-control and retrieval logic. Treating question difficulty as a nuisance factor—rather than letting it contaminate the temperature comparison—is the central motivation for the RCB design.

Questions were categorized into three blocks:

- **S — Simple:** Factual questions with a clear, direct answer in the rulebook (e.g., “How do you win?”, “How many players can play?”). 10 questions.
- **M — Misleading:** Questions that imply a rule exists that does not (e.g., “Is the player who holds the victim card awarded points?”). The correct response is a factual clarification or a polite correction. 10 questions.
- **F — Fallacious / Out-of-Scope:** Questions outside the rulebook entirely (e.g., “Why did the designers choose these suspects?”, “Where is Mr. Peacock?”). The correct response

⁶ PromptLayer, “LLM Temperature Setting.”

is an explicit refusal to answer. 11 questions.

By balancing all three blocks equally across temperature levels, question difficulty is separated from the temperature effect in the ANOVA model.

2.3 Factors Held Constant

To isolate the temperature effect, the following factors were controlled throughout the experiment:

- Model: claude-haiku-4-5-20251001 (fixed for all 93 runs)
 - Retrieval configuration: k = 3 chunks returned, BM25 + vector hybrid, contextual reranking
 - System prompt - a structured role-based constrained agent prompt with tool routing and conditional behavioral rules was identical for all runs
 - Chunking: semantic chunking by section header, 8 chunks total i.e a Retrieval Proportion of $\frac{3}{8} \sim 38\%$
 - Maximum tokens: fixed at the value specified in the project environment •
- Rulebook document: the 1986 Hasbro Clue Classic Detective Game Rules

2.4 Factor Summary Table

Temperature	Treatment (fixed)	3	0.3, 0.5, 0.8
Question Type	Blocking (fixed)	3 (S, M, F)	Simple, Misleading, Fallacious
Run Order	Nuisance (randomized)	93	Fully randomized across all runs
RAG Configuration	Held constant	N/A	Chunk size, retrieval k=3, reranker
API / Model	Held constant	N/A	claude-haiku-4-5-2025 1001

2.5 Design Structure and Randomization

The experiment followed a Randomized Complete Block Design (RCBD). The full design matrix of 93 runs (31 questions $\times$ 3 temperatures) was constructed upfront and then completely shuffled before execution. This complete randomization ensures that any drift in API performance, token-rate limits, or retriever state over the 13-minute execution window cannot systematically bias results toward one temperature level.

Each question appeared exactly three times—once at each temperature—making every question its own within-block control. This removes between-question variance from the treatment comparison and materially increases statistical power relative to a fully randomized design with the same total run count.

3. Response Variable

3.1 Rule Grounding Score (RGS)

The response variable is the Rule Grounding Score (RGS), a gated composite metric that combines correctness with evidence of grounding in the retrieved source material. The metric is designed to reward answers that are not only factually correct but also traceable to the rulebook, reflecting the core purpose of a RAG agent.⁷

RGS is defined as:

$$\text{RGS} = C_1 \times \frac{C_2 + C_3 + C_4 + C_5}{4}$$

Where:

$$C_1 \in \{0,1\}$$
$$C_2, C_3, C_4, C_5 \in \{0,1\}$$

C1 acts as a gate: if the answer is fundamentally wrong (or if the agent answers when it should have refused), the score is zero regardless of how well it cites its sources. If C1 = 1, the remaining four criteria contribute equally to determine how complete and well-grounded the correct answer is.

C1	0 or 1 (gate)	Correctness: answer matches the official rules, or agent correctly refuses an out-of-scope question
C2	0 or 1	Rule coverage completeness: all relevant rule conditions are included in the answer
C3	0 or 1	Citation presence: response references at least one source chunk (e.g., "According to CHUNK_X")
C4	0 or 1	Citation validity: the cited chunk actually supports the claim made
C5	0 or 1	Clarity: the response is clearly worded and directly addresses the question asked
RGS	[0, 1]	$C_1 \times (C_2 + C_3 + C_4 + C_5) / 4$ A score of 0.0 = incorrect or hallucinated. A score of 1.0 = fully correct and grounded.

3.2 Score Range and Interpretation

- **0.0:** Incorrect answer, hallucination, or failure to refuse an out-of-scope question. The gate criterion (C1) was not met.
- **0.25:** Correct but poorly grounded (e.g., right answer, no citation, partial coverage).
- **0.50 – 0.75:** Correct with partial grounding (some citation and coverage present).

⁷ Scoring is based on the utility of the answer and was inspired by Wei et al., "Tuning LLMs by RAG Principles."

1.0: Fully correct, fully cited, complete coverage, clearly expressed.

The response variable is continuous on [0, 1]. Individual question scores are effectively categorical, but the blocked ANOVA is applied to the residuals rather than the raw scores. Replication and balanced design support approximate normality and variance homogeneity in the residuals.

3.3 Measurement Consistency

Scoring was conducted manually after experiment completion using a fixed rubric. The next iteration will implement LLM as a judge validated by multiple reviewers. Each response was scored independently for C1 through C5 by a single rater using the definitions in the table above. To maintain calibration, all 93 responses were scored in block order (S, M, F) so that within-block consistency could be checked. Any borderline cases were resolved by re-reading the relevant rulebook chunk and applying the most conservative interpretation.

The RGS formula was applied programmatically after scoring to eliminate arithmetic error. No partial credit was assigned to individual criteria; all C values are strictly binary (0 or 1).

4. Results

4.1 Hypotheses

H₀: Mean RGS is the same across all three temperature levels (T = 0.3, 0.5, 0.8). Temperature has no statistically significant effect on rule-grounded response quality.

H₁: At least one temperature level produces a different mean RGS.

4.2 ANOVA Results

A one-way ANOVA with blocking on question type was performed in JMP. The model was:

$$Y_{ij} = \mu + \tau_i + \beta_j + \epsilon_{ij}$$

Temperature (Treatment)	2	0.006048	0.0654	0.9367
Question Type (Block)*	2	—	—	0.2254
Error	90	0.092428	—	—
C. Total	92	—	—	—

* Question Type (blocking factor) tested via REML variance component estimation (Wald p-value). The negative variance ratio (-0.018) rounds to zero, indicating no additional variance explained by blocking beyond the temperature treatment.

JMP Output



Note Consistency of response across all temperatures

Conclusion: Fail to Reject H_0

Temperature effect: $F(2, 90) = 0.0654$, $p = 0.9367$

The F-statistic of 0.0654 is far below the critical value and the p-value of 0.9367 provides no basis to reject the null hypothesis. The data do not support the claim that temperature affects mean rule-grounded response quality within the range tested ($T = 0.3$ to $T = 0.8$).

The blocking factor (Question Type) was also not statistically significant ($p = 0.2254$), indicating that question category (S, M, F) did not produce meaningfully different mean RGS values after accounting for the temperature treatment.

4.3 Discussion

The null result is practically informative. The F-statistic of 0.0654 is strikingly small—not merely below the significance threshold but near zero—indicating that virtually none of the observed variability in RGS scores can be attributed to temperature. This suggests that for a well-designed RAG agent operating within a narrow, rule-bounded domain, the temperature parameter does not materially degrade response quality across the range tested. The system prompt, retrieval configuration, and scope constraints appear to absorb the additional randomness introduced at higher temperatures, leaving the output quality essentially unchanged⁸.

The non-significance of the blocking factor warrants further investigation. One possible explanation is that the agent's out-of-scope refusal logic was sufficiently consistent to make F-type questions no harder than S-type questions in terms of RGS. Alternatively, retrieval errors at lower temperatures may have introduced noise that partially offset the expected advantage of more deterministic sampling. To test this groundedness, and Safety Aligned (Boundary) Performance Refusal, and instruction following metrics will be included in future experiments.

⁸ These results are congruent with Renze and Guven, who tested the effect of temperature on LLM output